MID 1
Subject:CS

Question 1

Q1: Define Authentication, Authorization, and Accounting (AAA) (2 marks)

Authentication: Verifying the identity of a user or device.
Example: Logging in with username and password.

Authorization: Determining what an authenticated user can do.
Example: Student can view grades but cannot edit them.

Accounting: Recording user activities and resource usage.
Example: Logging login times and accessed files.

OR

Q2: Define CIA Triad (2 marks)

1.Confidentiality: Ensures information is accessible only to authorized users.

Example: Private emails or medical records.

2.Integrity: Ensures data is accurate and not altered by unauthorized users.

Example: Bank transactions must remain correct.

3.Availability: Ensures information is available when needed.

Example: Online services like Google Drive must be accessible anytime.

Q3: Explain Vulnerability, Threat, and Risk with examples (4 Marks)

In information security, the terms Vulnerability, Threat, and Risk are closely related but have different meanings.

1) Vulnerability
A vulnerability is a weakness or flaw in a computer system, network, or software that can be exploited by an attacker. It may exist due to poor security practices, software bugs, or lack of updates.

Examples:
Using a weak password like “12345”
Not updating the operating system

Unsecured Wi-Fi network
Software bugs
A vulnerability itself does not cause damage, but it creates an opportunity for an attack.

2) Threat
A threat is any potential danger that can harm a system or data by exploiting a vulnerability. It can be intentional or accidental.

Examples:
Hackers trying to steal data
Malware or virus attacks
Phishing emails
Natural disasters like fire or flood
A threat represents the possible source of harm.

3) Risk
A risk is the possibility of loss, damage, or harm when a threat successfully exploits a vulnerability. It depends on both the likelihood of the threat and the impact of the damage.

Examples:
If a user has a weak password (vulnerability), a hacker (threat) may access the account → Data theft (risk).
If antivirus software is not updated (vulnerability), malware (threat) may infect the system → Data loss (risk).

OR

Q4: Explain the Importance of Confidentiality and Integrity with examples (4 Marks)

In information security, Confidentiality and Integrity are two important principles of the CIA triad (Confidentiality, Integrity, Availability). They help in protecting data from unauthorized access and modification.

1) Confidentiality
Confidentiality means protecting information from unauthorized access. Only authorized users should be able to view or access the data.

Importance:
Protects personal and sensitive information.
Prevents data leakage and misuse.
Maintains privacy of users and organizations.
Builds trust between customers and companies.

Examples:

Using passwords to protect email accounts.
Encrypting bank transaction details.
OTP (One Time Password) for online banking.
Access control in company systems (only employees can access internal data).
If confidentiality is not maintained, sensitive data like passwords, bank details, or exam papers may be leaked.

2) Integrity
Integrity means ensuring that data is accurate, complete, and not altered by unauthorized persons.

Importance:
Ensures data accuracy and reliability.
Prevents unauthorized modification of information.
Maintains correct records in business and banking.
Helps in making correct decisions based on accurate data.

Examples:

Digital signatures to verify documents.
Hash values to check file integrity.
Bank balance should not change without valid transaction.
Student marks should not be modified illegally.
If integrity is not maintained, wrong or modified data can cause financial loss or serious problems.

Q5: Describe any two layers of Cyber Security and their role (2 marks)

1. Network Security:
Network security protects the network infrastructure from unauthorized access or attacks.
It ensures safe communication between devices and systems.
Network security helps prevent malware, viruses, and cyber intrusions.

Example: Firewalls block unauthorized access to a network.

2. Application Security:
Application security protects software applications from threats and vulnerabilities.
It ensures applications work safely and data remains secure.
Application security prevents attacks like hacking, malware, and data leaks.

Example: Using secure coding practices to prevent SQL injection.

OR

Q6: Explain the need of Private Proxy in modern network security (2 marks)

Private Proxy:

A private proxy hides the user's real IP address, giving online anonymity.
It protects personal identity and sensitive data from hackers or trackers.
Allows safe access to blocked or restricted websites.
Helps prevent unauthorized access and unwanted monitoring of internet activity.

Example: Browsing safely on public Wi-Fi without revealing your IP.
Example: Companies using it to secure internal data when employees access the internet remotely.

Question 2

Any two (1,2 and 3)

Q1: Define Virus and Worm (2 marks)

1.Virus:
A virus is a malicious program that attaches itself to files or programs.
It spreads when the infected files are shared or executed.
Example: A virus in a downloaded software file.

2.Worm:
A worm is a standalone malware that spreads automatically across networks.
It does not need to attach to files to replicate.
Example: Internet worm spreading through email attachments.

Q2: Define Trojan Horse and Ransomware (2 marks)

1.Trojan Horse:
A program that looks legitimate but contains hidden malicious code.
Used to steal data, damage files, or create backdoors.

Example: Fake software that installs a keylogger.

2.Ransomware:
Malware that locks or encrypts user files and demands payment to unlock them.
Prevents access to important data until ransom is paid.

Example: WannaCry ransomware encrypting files and asking for Bitcoin payment.

Q3: Define Malware and list four types (2 marks)

Malware:

Malware is malicious software designed to harm or exploit computer systems.

Four types of malware:

1. Virus
2. Worm
3. Trojan Horse
4. Ransomware

Q4: Explain Buffer Overflow and Privilege Escalation (4 Marks)

In cybersecurity, Buffer Overflow and Privilege Escalation are common types of security attacks that can compromise a system.

1) Buffer Overflow

A buffer overflow occurs when a program stores more data in a memory buffer than it can handle.
When the buffer exceeds its limit, extra data overwrites adjacent memory, which may cause the system to crash or allow attackers to execute malicious code.

Causes:

Poor programming practices
Lack of input validation
Writing beyond allocated memory size

Examples:

A login form that allows unlimited characters in the username field.

An attacker enters very long input, which crashes the program.

In serious cases, the attacker can insert malicious code and gain control of the system.

Effects:

System crash
Data corruption
Unauthorized access

2) Privilege Escalation

Privilege escalation is an attack where a user gains higher access rights than they are allowed.

It happens when attackers exploit system vulnerabilities to move from a normal user account to an administrator (root) account.

Types:
Vertical Privilege Escalation – Gaining higher-level permissions (user → admin).
Horizontal Privilege Escalation – Accessing another user's account with same permission level.

Examples:
A normal user exploits a software bug to get administrator rights.
A student accesses another student's account without permission.

Effects:
Full system control by attacker
Data theft or deletion
Installation of malware

OR

Q5: Explain Security Role of OSI Model in Preventing Attacks (4Marks)

The OSI (Open Systems Interconnection) model divides network communication into 7 layers. Each layer has a specific role, and security measures can be applied at different layers to prevent attacks.

Security Role of Each OSI Layer

1) Physical Layer
Protects physical devices like cables and routers.
Security: Locked server rooms, CCTV, access control.
Prevents: Physical tampering and cable theft.

2) Data Link Layer
Controls communication between devices in the same network.
Security: MAC address filtering.
Prevents: Unauthorized device connection.

3) Network Layer
Handles routing and IP addressing.
Security: Firewalls and IP filtering.
Prevents: IP spoofing and unauthorized access.

4) Transport Layer
Ensures reliable data transfer.
Security: SSL/TLS encryption.

Prevents: Data interception and session hijacking.

5) Session Layer
Manages sessions between users.
Security: Session timeout and authentication.
Prevents: Session hijacking.

6) Presentation Layer
Converts and encrypts data.
Security: Data encryption and decryption.
Prevents: Data leakage.

7) Application Layer
Provides services like email, web browsing.
Security: Antivirus, authentication, access control.
Prevents: Malware and phishing attacks.

Q6: Explain any three OT (Operational Technology) attacks (3 marks)

Stuxnet Attack:
A malware attack on industrial control systems in nuclear plants.
It manipulated machinery operations and caused physical damage to centrifuges.
Showed that cyber attacks can have real-world physical effects.

Triton/Trisis Attack:
Targeted safety systems in industrial facilities.
Could disable safety shutdowns, putting human life and equipment at risk.
Highlighted the threat to critical safety systems.

BlackEnergy Attack:
Focused on power grids to disrupt electricity supply.
Led to temporary blackouts and operational failures.
Demonstrated how cyber attacks can affect essential infrastructure.

OR

Q7: Explain any three IoT attacks with examples (3 marks)

Botnet Attack (Mirai Botnet):
Compromised IoT devices like cameras and routers.
Used them to perform large-scale DDoS attacks.
Example: Mirai Botnet caused major internet outages in 2016.

Data Theft:

Hackers access sensitive data from IoT devices.
Personal and private information can be stolen.
Example: Smart home cameras leaking private videos.

Device Hijacking:
Attackers take control of IoT devices remotely.
Can misuse devices for malicious purposes.
Example: Hackers controlling smart locks or thermostats without permission.

Question 3

Any two (1,2 and 3)

Q1: Define Cybercrime (2 marks)

Cybercrime is any illegal activity carried out using computers, networks, or the internet.
It includes crimes that target people, property, organizations, or society.
Cybercrime involves stealing, altering, or destroying digital information.
Example: Hacking someone's bank account to steal money.

Q2: Types of Cyber Criminals (2 marks)

Hackers: Gain unauthorized access to systems to steal or damage data.

Phishers: Trick people into revealing sensitive information like passwords or bank details.

Cyber Terrorists: Use cyber attacks to create fear, disrupt services, or cause political/economic damage.

Insiders: Employees or trusted users who misuse their access to harm the organization.

Script Kiddies: Unskilled users who use pre-written hacking tools or scripts to attack systems.

Q3: Cyber Bullying and Phishing (2 marks)

Cyber Bullying:
Using digital devices like computers, phones, or social media to harass, threaten, or embarrass someone.
Can cause emotional or psychological harm to victims.
Example: Sending harmful or threatening messages on social media.
Example: Posting embarrassing photos or videos online without consent.

Phishing:
Fraudulent attempt to obtain sensitive information like passwords or credit card numbers.

Usually done by pretending to be a trusted person or organization.
Example: Fake emails from “banks” asking for login credentials.
Example: Scam websites mimicking real websites to steal user data.

Q4: Classification of Cyber Crimes based on target categories (4 marks)

Cybercrime against individuals:
Targets personal data, privacy, or reputation.
Can harm emotions, finances, or personal security.
Example: Identity theft, cyber bullying, or harassment online.

Cybercrime against property:
Targets digital or financial property.
Can involve stealing money, files, or digital assets.
Example: Online fraud, ransomware, or data theft.

Cybercrime against organizations:
Targets businesses, institutions, or government systems.
Can disrupt operations, steal sensitive information, or damage reputation.
Example: Hacking company servers, stealing trade secrets.

Cybercrime against society:
Affects public safety, social order, or large groups of people.
Can create panic, misinformation, or mass damage.
Example: Spreading fake news, cyber terrorism, or large-scale malware attacks.

OR

Q5: Cybercrimes against organizations with examples (4 marks)

Hacking:
Unauthorized access to company systems or networks.
Can steal sensitive data or disrupt operations.
Example: Hackers stealing customer information from a corporate server.

Insider Threats:
Employees or trusted users misuse their access to harm the organization.
Can leak confidential information or sabotage systems.
Example: Employee copying trade secrets for personal gain.

Ransomware Attacks:
Malware that encrypts organizational data and demands ransom.
Can block access to important files and disrupt business operations.
Example: Hospitals hit by ransomware, unable to access patient records.

Denial of Service (DoS) Attacks:
Flooding the organization's network or website to make it unavailable.
Can interrupt services and damage reputation.
Example: Online banking websites becoming inaccessible due to DDoS attack.

Q6: Preventive measures to reduce cybercrime (3 marks)

1.Use Strong Passwords
Create strong and unique passwords. Use a mix of letters, numbers, and symbols. Change passwords regularly.

2.Install Antivirus and Firewall
Use updated antivirus software and enable firewall protection to prevent malware and unauthorized access.

3.Regular Software Updates
Keep operating systems and applications updated to fix security vulnerabilities.

4.Be Aware of Phishing Emails
Do not click on unknown links or download attachments from suspicious emails.

5.Data Backup
Take regular backups of important data to avoid loss during cyber attacks.

OR

Q7: Major cyber offences under IT Act 2008 (Sec 65, 66, 67) – 3 marks

Section 65 – Tampering with computer source code:

Illegal to knowingly destroy, alter, or conceal computer source code.
Protects the integrity of software and programs.

Example: Modifying software code to hide a virus or malware.

Section 66 – Hacking:

Unauthorized access to computer systems or networks.
Can lead to data theft, disruption, or damage to systems.

Example: Breaking into a bank's server to steal money or sensitive information.

Section 67 – Publishing or transmitting obscene material:

Sending, publishing, or distributing obscene or offensive content electronically.
Aims to prevent misuse of digital platforms for immoral purposes.

Example: Sharing adult content on social media or websites.